



The Electrifying Landscape Of EV CHARGING SYSTEMS

Wireless charging, swappable charging, robotic charging—many different approaches have emerged to charge EVs. Why? How are they faring? Which ones will lead?



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In recent years, the global electric vehicle (EV) industry has witnessed an unprecedented surge, marking a remarkable transformation in the automotive landscape. “From approximately 3.1 million EV sales in 2020, the figures [rocketed] to over 9 million in 2022. Predictions highlight the rapid growth, estimating 12 million EV sales in 2023, possibly [reaching] 20 million by 2025,” says Shazan Siddiqi, Technology Analyst, IDTechEX. The surge isn’t just in passenger vehicles. Commercial EV sales have also seen a notable increase, with leading truck manufacturers expecting electric trucks to make up a significant share, anywhere from 35% to 60%, of their annual sales by 2030.

As the EV market evolves, charging infrastructure follows suit, expanding rapidly to accommodate the burgeoning demand. In 2022, global public charging points reached nearly 2.7 million, with almost a

million chargers installed within the same year. “The growth is ready to accelerate, with estimates suggesting that a staggering 222 million charging events will be needed by 2034 to support the expanding EV fleet,” adds Shazan. Correspondingly, investments in charging infrastructure are on the rise, with a projected cumulative global investment exceeding \$123 billion by 2034. Amid this dynamic landscape, the focus on establishing robust global DC fast-charging networks takes centre stage.

From traditional charging systems to innovative alternatives, the landscape is evolving at an unprecedented rate, shaping the future of transportation. While the year 2023 promises to be a pivotal period for EVs and their charging ecosystem, the narrative extends beyond mere statistics. The surge in EV adoption isn’t just changing how we drive, it’s redefining the infrastructure that powers our journey